

HOAN NGUYEN

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RESEARCH INTERESTS

Aubry-Mather theory, Minimal surfaces, Min-Max theory, Geometric flows.

EDUCATION

The University of Chicago

- Ph.D. in Mathematics (expected graduation: 2027).
- Ph.D. Advisor: André Neves.

The University of Chicago

- Master's in Mathematics (2022).

Hong Kong University of Science and Technology

- B.Sc. in Mathematics (2021).
- Senior thesis: *Kähler-Ricci flow and conformal submersion*.
- Thesis Advisor: Frederick Tsz-Ho Fong.

PUBLICATIONS AND PREPRINTS

Differential Geometry and Geometric Analysis

- **A min-max gap characterization of minimal foliations on the torus.**
Preprint, 2026. <https://arxiv.org/abs/2605.12428>
- **Kähler-Ricci flow and conformal submersion.**
Mediterranean Journal of Mathematics, 2023.

Other Publications

- **Residual entropy of confined and interfacial monolayer water ice.**
Joint with Lien Le. *Computational and Theoretical Chemistry*, 2026.

INVITED TALKS

1. *A min-max gap characterization of minimal foliations on a torus and applications*. Urbana-Champaign, UIUC Geometric Analysis Seminar, Apr 21, 2026.

¹Updated May 13, 2026

TEACHING EXPERIENCE

The University of Chicago

- Instructor: Elementary Functions and Calculus I,II,III, Calculus I, II, Linear Algebra.
- Teaching Assistant: Accelerated Analysis, Introduction to Dynamical Systems, Functional Analysis.

OTHER SKILLS

- Python, R (basic)

AWARDS

- Gold medals at the International Mathematical Olympiad (2014, 2015).
- HKUST Academic Excellence Award.